264. Ugly Number II

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Write a program to find the `n` -th ugly number.

Ugly numbers are **positive numbers** whose prime factors only include `2, 3, 5` .

Example:

```
Input: n = 10
Output: 12
Explanation: 1, 2, 3, 4, 5, 6, 8, 9, 10, 12 is the sequence of the first
10 ugly numbers.
```

Note:

1. `1` is typically treated as an ugly number.
2. `n` **does not exceed 1690**.

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Hide Hint 1

The naive approach is to call `isUgly` for every number until you reach the nth one. Most numbers are *not* ugly. Try to focus your effort on generating only the ugly ones.

Show Hint 2

Show Hint 3

Show Hint 4

Java Autocomplete

```java

    public class Solution {
    public int nthUglyNumber(int n) {
        int[] ugly = new int[n];
        ugly[0] = 1;
        int index2 = 0, index3 = 0, index5 = 0;
        int factor2 = 2, factor3 = 3, factor5 = 5;
        for(int i=1;i<n;i++){
            int min = Math.min(Math.min(factor2,factor3),factor5);
            ugly[i] = min;
            if(factor2 == min)
                factor2 = 2*ugly[++index2];
            if(factor3 == min)
                factor3 = 3*ugly[++index3];
            if(factor5 == min)
                factor5 = 5*ugly[++index5];
        }
        return ugly[n-1];
    }
}

```

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